

Control Systems Mid Exam 2023-2024

تم حل هذا الملف من قبل الطالب : ليث عمرو

هذا الحل هو جهد بشري قابل للأخطاء ،
إن أحسنت فمن الله وإن أسأت فمني ومن الشيطان .

لا تنسوا إخواننا المسلمين عامة وفي فلسطين وغزة هاشم خاصة من صالح دعاءكم .

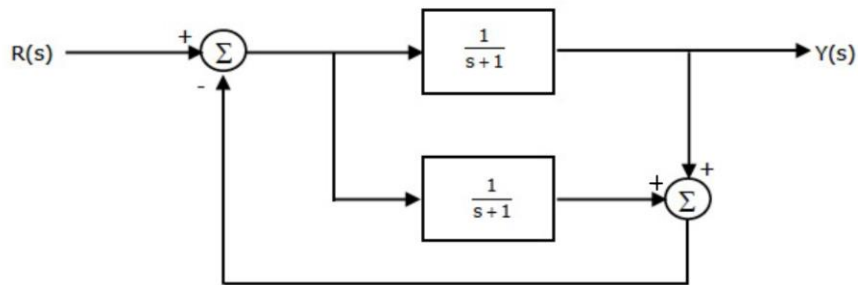
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Question 1) answer the following question

10 marks

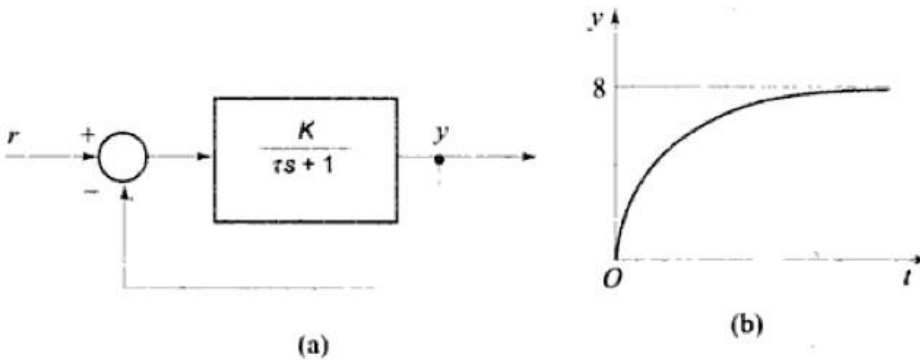
1. The transfer function from $R(s)$ to $Y(s)$ is :



$$Y(s)/R(s) = \frac{2}{s+3}$$

2. The time response of the system shown below in (a) to an input $r(t) = 10, t > 0$ is shown in (b).

The gain K is equal to



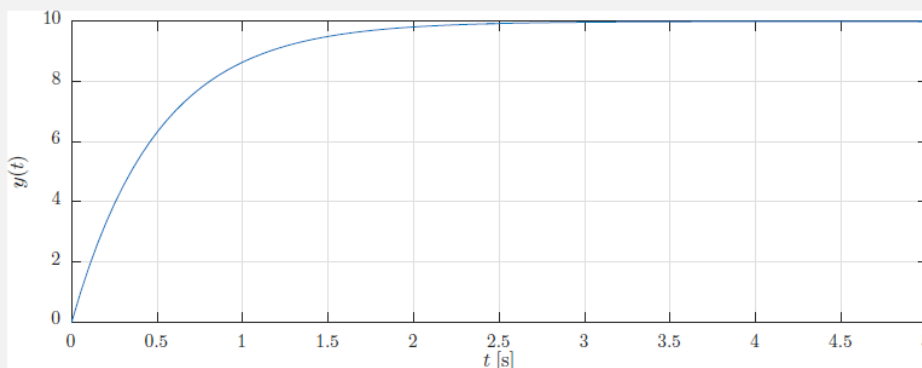
$K =$

0.8

3. The figure below shows the step response of a first-order system of the form

$$\dot{x}(t) = a \cdot x(t) + b \cdot u(t)$$

$$y(t) = x(t) .$$



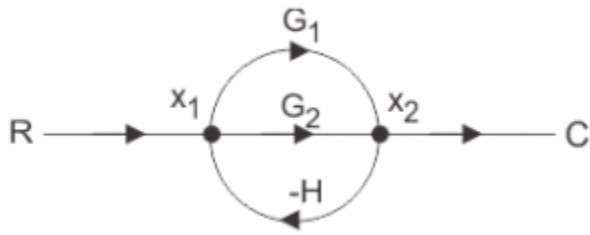
$a =$ **-2**

$b =$ **20**

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4. Use mason's gain formula to find the transfer function of the given figure:



$$C/R = \frac{G1+G2}{1+G1H+G2H}$$

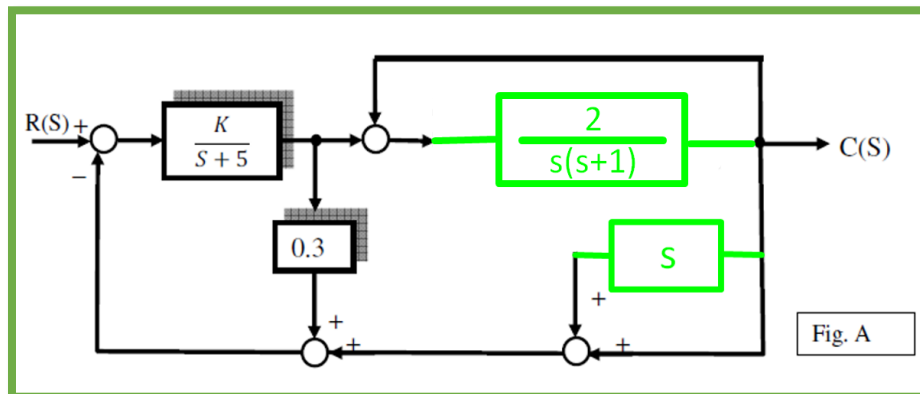
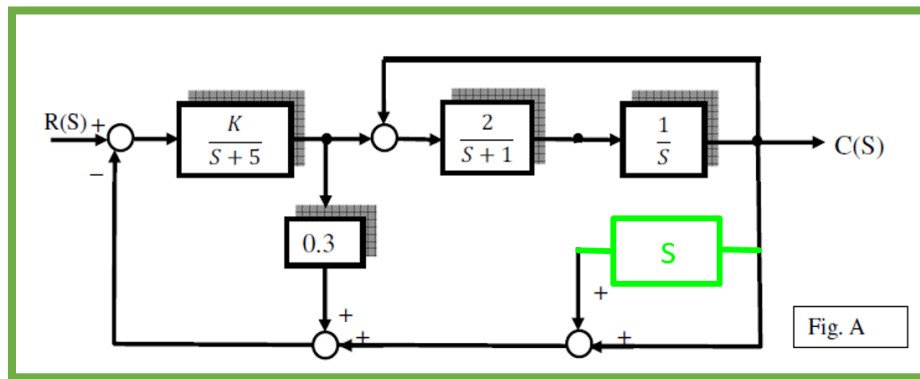
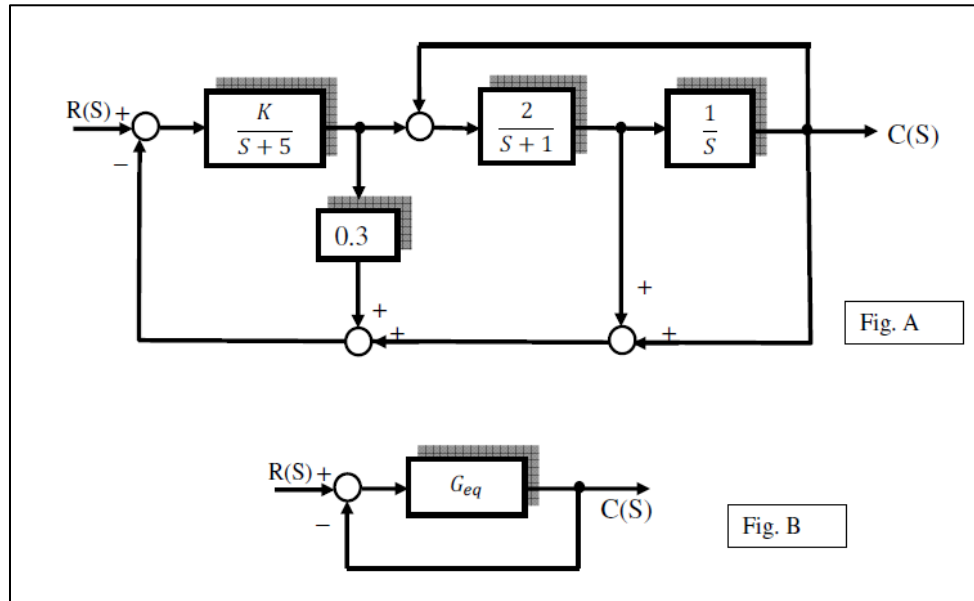
5. A system is given by its transfer function. $g(S) = \frac{s-1}{s^2+s-2}$ Which one of the following state space representations is equivalent to the system given by its transfer function?

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

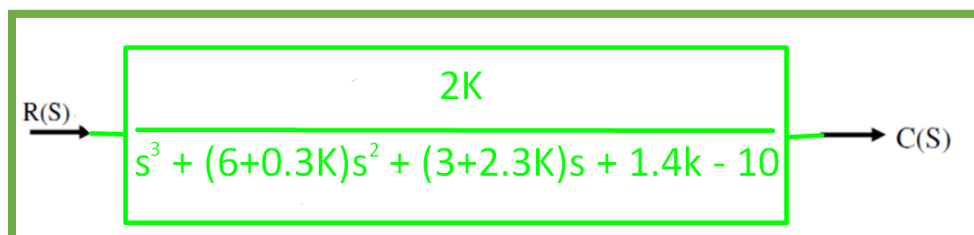
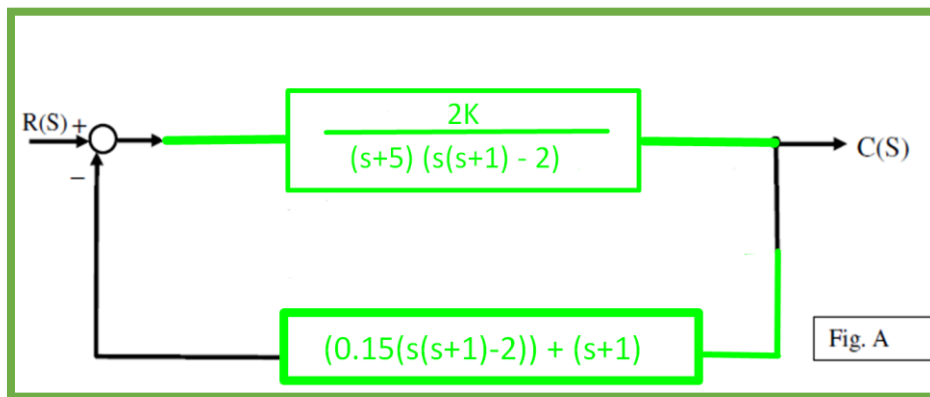
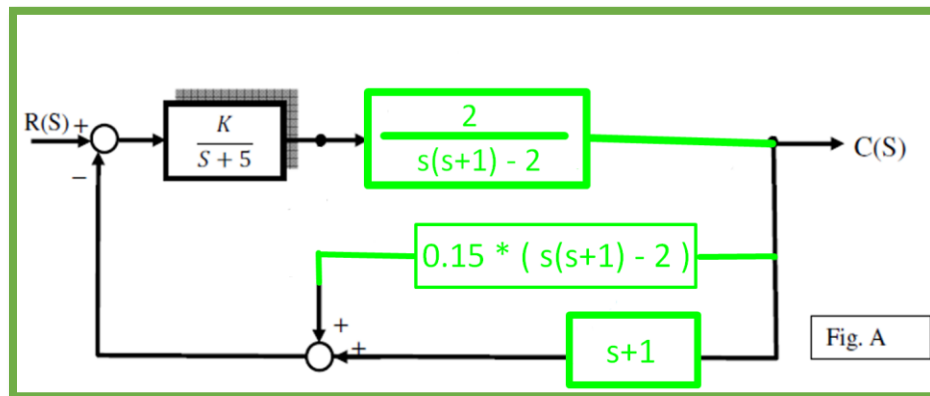
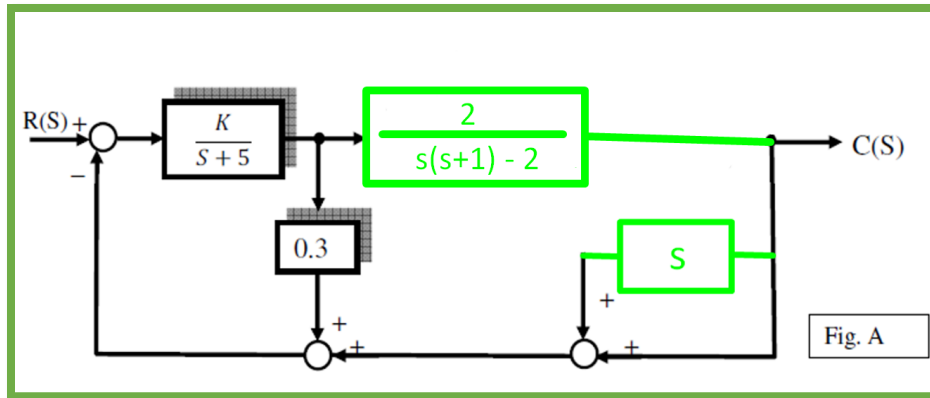
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Question 2) If the control systems shown in Fig. A and B are equivalent, Find Geq. (10marks)



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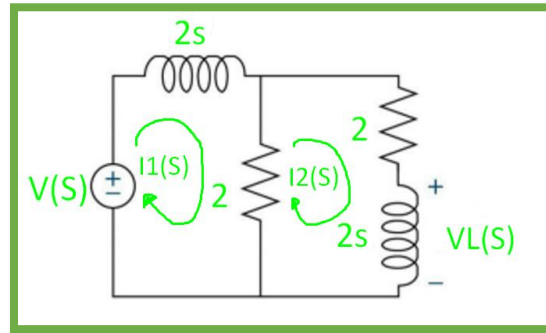
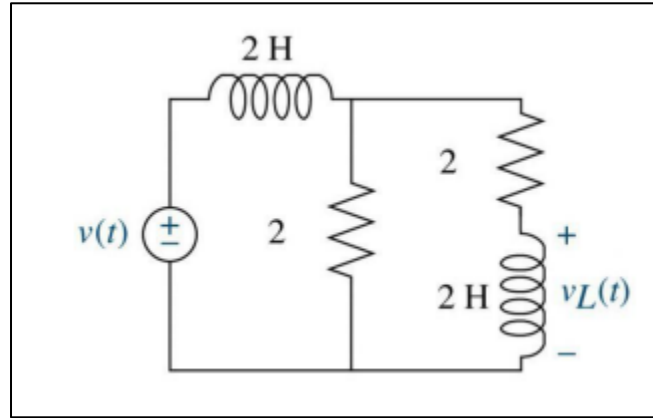
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Question 3) for the electrical system shown below find the transfer function of the system $V_L(s)/v(s)$ **10 marks**



$$(2s+2) I_1(S) - 2 I_2(S) = V(S) \dots \text{Eq.(1)}$$

$$(2s+4) I_2(S) - 2 I_1(S) = 0 \dots \text{Eq.(2)}$$

$$V_L(S) = 2s * I_2(S) \rightarrow I_2(S) = \frac{V_L(S)}{2s}$$

$$\text{From Eq.(2)} \rightarrow I_1(S) = (s+2) * I_2(S) \rightarrow I_1(S) = (s+2) * \frac{V_L(S)}{2s}$$

Substitute in $I_1(S)$ and $I_2(S)$ in Eq.(1) :

$$\frac{(2s+2) [(s+2) * V_L(S)]}{2s} - 2 * \frac{V_L(S)}{2s} = V(S)$$

$$\left[s + 3 + \frac{2}{s} \right] V_L(S) - \frac{V_L(S)}{s} = V(S)$$

$$V_L(S) * \left[\frac{s^2 + 3s + 2}{s} - \frac{1}{s} \right] = V(S)$$

$$\frac{V_L(S)}{V(S)} = \frac{s}{s^2 + 3s + 1}$$

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